Ball and Players Tracking in Padel Matches Videos

Andrea Cauda
Politecnico di Torino
Turin, Italy
s343386@studenti.polito.it

Davide Tonetti
Politecnico di torino
Turin, Italy
s334297@studenti.polito.it

Antonio Visciglia
Politecnico di torino
Turin, Italy
s346837@studenti.polito.it

Abstract

Padel is one of the fastest-growing racket sports worldwide, characterized by high-speed rallies, complex player interactions, and constrained playing spaces. Despite this complexity, performance analysis in padel still largely relies on manual video review, which is time-consuming, subjective, and difficult to scale. In this work, we propose an automated, data-driven framework for player and ball tracking in padel match videos using deep learning-based object detection. Leveraging a large-scale, frame-annotated dataset captured from a single fixed camera, we evaluate multiple YOLO-based architectures and analyze the impact of input resolution on detection accuracy and inference efficiency. Beyond detection, we introduce a pipeline for extracting actionable performance statistics, including player heatmaps and average inter-player distance, enabling objective tactical insights. Experimental results show strong performance in player detection and highlight current limitations in ball recall, motivating future fine-tuning and tracking-based refinements. This work establishes a reproducible baseline for automated padel analytics and demonstrates the potential of computer vision to support coaches, players, and performance analysts. The source code is available at: P10 Ball and Players Tracking in Padel Matches Videos.

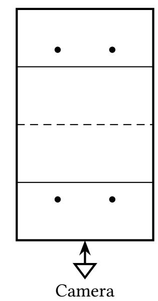

Figure 1: Schematic single fixed-camera setup and court layout used in this work (top-view, not to scale).

1 Introduction

Padel has rapidly evolved into a globally popular sport, combining elements of tennis and squash while introducing unique tactical and spatial dynamics. Matches are characterized by frequent changes of direction, close player proximity, glass-wall interactions, and fast ball trajectories. These factors make padel particularly challenging to analyze using traditional, manual video-based methods and currently, coaches and performance analysts rely heavily on manual video review to evaluate player positioning, movement patterns, and tactical behavior. While this approach can provide valuable qualitative insights, it suffers from several limitations: it is time-consuming, subjective, and difficult to scale across matches, players, or training sessions. Moreover, extracting quantitative metrics such as spatial coverage, player distance, or movement density often requires extensive manual annotation or is not feasible at all. In recent years, computer vision-based sports analytics has gained increasing relevance. Industrial systems such as **Amazon Prime Vision** [1] demonstrate how real-time tracking and visual overlays can enrich sports analysis and broadcasting by transforming raw video into actionable insights. However, these solutions are typically designed for high-budget, multi-camera environments and are not easily transferable to sports contexts with more limited sensing setups, such as padel. Despite its growing popularity, padel remains underexplored in the scientific literature: the sport presents several technical challenges for automated analysis, including frequent occlusions, reflections from glass walls, rapid ball motion, and the small size of the ball relative to the playing area. These factors complicate reliable detection and tracking, particularly in single-camera scenarios. In this paper, we present a data-driven framework for automated player and ball detection in padel match videos, designed for a single fixed-camera setup (see fig:padel-setup).

Using the **PadelTracker100** dataset [2], consisting of approximately 100,000 annotated frames, we evaluate lightweight **YOLO-based** [3] detection models under different input resolutions, analyzing the trade-off between precision, recall, and inference time. Beyond detection, we propose a pipeline for extracting interpretable performance statistics, such as player heatmaps and average inter-player distance, enabling objective tactical analysis. The contributions of this work are threefold: (1) an experimental evaluation of lightweight object detection models for padel-specific player and ball tracking; (2) a practical pipeline for transforming detection outputs into meaningful performance statistics; and (3) an analysis of current limitations and future directions toward scalable, automated padel analytics.

2 Related Work

Recent advances in computer vision have significantly improved automated analysis of sports videos, particularly through deep learning–based object detection and tracking approaches. A common paradigm adopted in the literature is tracking-by-detection, where objects are detected independently in each frame and subsequently associated over time to reconstruct trajectories.

Player detection and tracking have been extensively studied in team and racket sports, demonstrating that convolutional neural networks can achieve robust localization in dynamic and crowded scenes. In contrast, ball detection remains a challenging task due to the small size of the object, fast motion, and frequent occlusions. Prior work on racket sports highlights the need for higher input resolutions, temporal smoothing, or motion-based constraints to improve robustness in such scenarios.

Single-stage object detectors from the YOLO family are widely used in applied sports analytics due to their efficiency and competitive accuracy. Lightweight YOLO variants are particularly suitable for video-based pipelines where inference speed and resource constraints are critical.

Beyond detection, several studies emphasize the importance of converting low-level detections into higher-level spatial and tactical representations. Heatmaps and distance-based metrics are commonly employed to summarize player positioning and interaction patterns, underscoring the dependency of analytical quality on detection accuracy and temporal consistency.

3 Method

3.1 Problem Statement

The goal of this work is to design an automated framework for *player and ball tracking in padel match videos*, enabling the extraction of objective and interpretable performance statistics. Considering a padel match video captured from a *single fixed camera*, conventional analysis is typically performed through manual inspection, which limits the systematic extraction of quantitative and reproducible information. We therefore reformulate this task as a computer vision problem based on object detection and spatio-temporal aggregation.

Formally, let $V = \{I_t\}_{t=1}^{T}$ be a sequence of video frames extracted from a padel match, where $I_t \in \mathbb{R}^{H \times W \times 3}$. For each frame I_t, the objective is to detect a set of objects $O_t = \{o_t^k\}_{k=1}^{N_t}$, where each object o_t^k belongs to one of the predefined classes $c \in \{\text{player}, \text{ball}\}$.

Each detected object is represented by a bounding box

$$b_t^k = (x_t^k, y_t^k, w_t^k, h_t^k), \tag{1}$$

together with a confidence score $s_t^k \in [0, 1]$. The detection task can therefore be formulated as learning a function

$$f_\theta : I_t \longrightarrow \{(b_t^k, c_t^k, s_t^k)\}_{k=1}^{N_t}, \tag{2}$$

parameterized by θ, which maps each frame to a set of player and ball detections.

Beyond frame-level detection, the objective is to derive higher-level performance indicators by aggregating detections across time. In particular, we focus on spatial density estimation of player positions (heatmaps) and distance-based metrics derived from player trajectories. This formulation enables a transition from raw video data to quantitative and reproducible padel analytics.

3.2 Computing Infrastructure

Table 1: Comparison between Asgard (LINKS Foundation) and Google Colab (free version).

Feature	Asgard (LINKS)	Google Colab (Free)
CPU	2× Intel Xeon Silver 4216 @ 2.20 GHz	Shared virtual CPU
System memory	377 GB RAM	Limited (∼12–25 GB RAM)
GPU availability	Dedicated GPUs	Shared GPUs
GPU model	8× NVIDIA RTX 2080 Ti	Varies (T4 / K80)
GPU memory	11.8 GB per GPU	Limited (typically ≤16 GB)
Disk storage	7.0 TB RAID10 (persistent)	Limited and temporary (∼15 GB)
Usable project storage	∼300 GB	Not guaranteed
Session duration	No enforced limits	Time-limited sessions
Runtime stability	High	Subject to interruptions
Data persistence	Persistent across sessions	Data may be cleared
Suitability for video data	High	Limited
Reproducibility	High	Limited

Due to the video-based nature of the task and the size of the dataset, substantial computational resources are required as the pipeline involves storage of raw videos and extracted frames, training and evaluation of deep neural networks, and long-running inference processes at multiple input resolutions.

As shown in Table 1, Google Colab presents critical limitations for video-based computer vision tasks, including restricted and non-persistent disk storage, shared computational resources, and session interruptions. These constraints make it impractical to store full-length match videos, extracted frames, and intermediate inference outputs, as well as to perform long and reproducible experiments.

In contrast, Asgard provides a stable high-performance computing environment with large persistent storage, dedicated GPUs, and substantial system memory 1. This infrastructure enables systematic experimentation, evaluation across multiple configurations, and uninterrupted training and inference workflows.

3.3 Method Overview and Pipeline

(1) **Video Preprocessing:** input match videos are decomposed into individual frames and resized according to the selected input resolution.
(2) **Object Detection:** each frame is processed by a deep learning–based object detector to identify players and the ball, producing bounding boxes, class labels, and confidence scores.
(3) **Post-processing and Filtering:** detections are filtered to remove low-confidence predictions and out-of-court objects, reducing noise introduced by background elements and reflections.
(4) **Temporal Aggregation:** frame-level detections are aggregated across time to form approximate player trajectories; isolated missing detections are mitigated through temporal continuity assumptions.
(5) **Statistical Analysis:** aggregated detections are projected onto a normalized court representation to compute player heatmaps and average inter-player distance metrics.

This modular pipeline, that can be seen in Figure 2, allows each component to be improved independently and facilitates systematic experimentation with different models and configurations.

3.4 Detection Models

For object detection, we adopt YOLO-based architectures [6–8], which provide an effective balance between detection accuracy and inference efficiency by performing detection in a single forward pass. Lightweight variants are considered to accommodate video-based processing constraints. Specific model configurations and comparative evaluations are reported in the Experimental Section.

model	parameters
yolo11n	2.6M
yolo10n	2.3M
yolo9t	2.0M

Table 2: Yolo model comparison

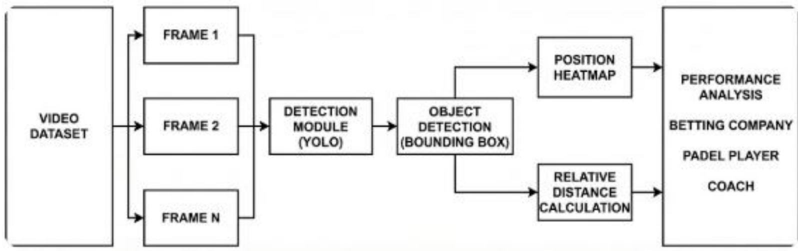

Figure 2: Overall pipeline of the proposed framework, from a sin- gle fixed-camera match video to detections, temporal aggregation, and derived performance statistics.

3.5 Input Resolution

Input frames are resized according to the **imgsz hyperparameter** [5], which defines the spatial resolution of the input images processed by the detector. Larger input resolutions improve detection accuracy for small objects such as the ball, at the cost of increased memory usage and inference time. Multiple resolutions are therefore considered to systematically study this trade-off.

3.6 Statistics

The transition from raw object detection to actionable performance metrics requires a robust post-processing and aggregation framework. This section details the methodology used to transform discrete frame-level bounding boxes into continuous trajectories and high-level tactical statistics.

3.6.1 From Detections to World Coordinates. The output of the YOLO-based detector provides bounding boxes in pixel coordinates, denoted as $b_t^k = (x_t^k, y_t^k, w_t^k, h_t^k)$, where (x, y) represents the top-left corner of the bounding box, while w and h represent its width and height, respectively; in order to be able to perform meaningful tactical analysis, these must be projected onto a normalized 2D court representation. We define the player's ground position P_t^k as the midpoint of the bounding box's bottom edge, approximating the contact point with the turf:

$$P_{t,\text{pixel}}^k = \left(x_t^k + \frac{w_t^k}{2}, y_t^k + h_t^k \right) \tag{3}$$

The transformation from the image plane to the world plane is achieved through a planar projective transformation, or homography. To compute the 3×3 homography matrix H, we perform a calibration process using eight known correspondence points between the image coordinates (u, v) and the physical world coordinates (x, y), using critical court landmarks.

For increased robustness against manual annotation errors in the pixel space, the matrix H is estimated using the *Random Sample Consensus* (RANSAC) [4] algorithm: this iterative method selects a subset of points to compute a candidate homography and evaluates it against the remaining points, effectively filtering outliers that exceed a reprojection error threshold of 5.0 pixels.

The resulting vector is normalized by w yields the final coordinates in meters relative to a court origin $(0, 0)$ located at the top-left corner of the playing surface. Finally, to ensure data purity, detections falling outside the legal court boundaries, extended by a 1-meter buffer to account for out-of-glass plays, are discarded to eliminate possible detections of figures in the public.

3.6.2 Player Tracking and Temporal Continuity. Given the high-speed nature of padel and frequent occlusions, independent frame detection is insufficient, therefore we implemented a tracking-by-detection paradigm using momentum-based association logic.

(1) **Initialization and Labeling:** The tracker remains dormant until a "clean" frame with exactly four valid player detections is found. At this stage, identities are assigned based on court quadrants, detections are first split by the net line into Team A and B, then within each team, players are sorted by their X-coordinates resulting in labels A1, A2 and B1, B2. This ensures a stable and reproducible starting configuration for the duration of the rally.

(2) **Greedy Matching:** For each subsequent frame t, we maintain a state for each player k consisting of position P and velocity V. The predicted position is calculated as:

$$\hat{P}_t^k = P_{t-1}^k + V_{t-1}^k \tag{4}$$

The system constructs a distance matrix D between all predicted positions $\hat{P}_t$ and new detections O_t using Euclidean distance. A greedy matching algorithm iteratively selects the pair (k, o) with the minimum distance, assigns the detection to the player identity, and removes both from the matrix until all possible matches are exhausted.

(3) **Handling Missing Detections:** In instances where a player identity k fails to find a matching detection, due to occlusion or detector miss, the system generates a "Ghost Point". The player's position is updated using the predicted $\hat{P}_t^k$, but the velocity is attenuated by a decay factor $\gamma = 0.95$, ensuring that the unanchored "ghost" will eventually converge to a halt, preventing non-physical linear drift across the court:

$$V_t^k = V_{t-1}^k \cdot \gamma \tag{5}$$

(4) **Video Cuts detection:** The pipeline's stability relies on the ability to tackle cuts in the database's videos and reset the tracking mechanism accordingly; to this end the Spatial Jump Threshold is set at ($\tau = 1.0$ m). Since at a 30 fps capture rate, a 1.0 m displacement between frames implies an instantaneous velocity of 30 m/s ($\approx$ 108 kph), well beyond human capabilities, any $\Delta P > \tau$ is categorized as a tracking artifact, typically caused by video cuts, necessitating a system reset.

3.6.3 *Ball Tracking and Occlusion Recovery.*

To keep the system consistent, we use the same tracking rules and math for the ball as we do for the players in order to "fill in the gaps" when the ball is missing in certain frames ,which is very important since our AI model only manages to find the ball about 58% of the time.

When the detector fails to register the ball, the system switches to a prediction-only mode, using the last known velocity vector to estimate current coordinates. For the ball, the velocity decay factor is tuned to $\gamma_{ball} = 0.98$, representing a more aggressive persistence compared to players.

A key distinction between the two tracking systems is the handling of video cuts: unlike the player tracker, the ball tracker does not utilize the $\tau = 1.0$ m spatial jump threshold for automatic resets given that the ball can naturally achieve velocities that approach or exceed this threshold in high-impact scenarios (e.g., smashes or glass bounces), and a reset based on player-oriented locomotion limits would introduce false disruptions in the ball's trajectory.

3.6.4 *Statistics Visualization.*

The final objective of the pipeline is the synthesis of filtered and aggregated trajectories into high-level tactical indicators.

- **Players Heatmaps:** To analyze court coverage, we generate spatial density estimates and world coordinates $P^k_{t,\text{world}}$ are aggregated into a 2D histogram. This heatmap visualizes the frequency of a player's presence in specific zones, allowing for the identification of tactical tendencies such as baseline defensive play versus net-rushing offensive behavior.
- **Average Inter-Player Distance:** To quantify the synchronization between partners (e.g., the "rubber band" effect), we calculate the Euclidean distance between teammates at each time step. For Team A, consisting of players P^{A1} and P^{A2}, the instantaneous distance d_t is:

$$d_t = \sqrt{(x_t^{A1} - x_t^{A2})^2 + (y_t^{A1} - y_t^{A2})^2} \tag{6}$$

  The system outputs the mean $\bar{d}$ across the rally, providing an objective metric for team cohesion and defensive positioning.
- **Bird's Eye View Reconstruction:** For qualitative review, the system generates an animated tactical representation of the match. By mapping all player and ball coordinates onto a scaled 2D court template, we produce a "Bird's Eye View" (BEV) reconstruction facilitating the analysis of spatial interactions that are often obscured by the perspective of the original camera feed.

Estimation Quality: To ensure the transparency of these metrics, we define an *Estimation Ratio* (E_r):

$$E_r = \frac{\text{Total Estimated Points}}{\text{Total Calculated Points}} \times 100 \tag{7}$$

This percentage indicates the framework's reliance on predictive "Ghost Points" versus raw detection data, serving as a confidence score for the derived analytics.

4 Experiment

4.1 Dataset

The experimental evaluation is conducted using the **PadelTracker100** dataset [2], a large-scale repository specifically designed for padel-specific computer vision tasks. The dataset consists of approximately 100,000 frames extracted from professional padel matches padel matches, including both male and female matches, which is critical for ensuring the generalization of the detection models; the frames are all captured using a single fixed camera setup as described in Section 1 and each one is manually annotated with bounding boxes for two distinct classes: players and the ball.

To perform the experiments, the dataset was partitioned into three subsets:

- **Training Set (70%):** Used to fine-tune the YOLO architectures, allowing the models to learn the specific features of the padel ball and player postures.
- **Validation Set (10%):** Employed during the training phase to monitor for overfitting and to tune hyperparameters such as the input resolution (*imgsz*).
- **Test Set (20%):** Reserved for the final evaluation of precision and recall reported in Section 4.3, ensuring that the performance metrics reflect the models' ability to handle unseen rallies.

Dataset Limitations and Challenges: A critical characteristic of the PadelTracker100 dataset is that it is derived from professional television broadcasts rather than a raw, uninterrupted single-camera feed. Consequently, the footage contains frequent "video cuts" where the broadcast switches from the primary top-view camera to close-ups, slow-motion replays, or alternative court angles.

These discontinuities present significant challenges for temporal aggregation:

- **Identity Loss:** Frequent cuts make it impossible to maintain player IDs throughout the entire duration of a match.
- **Team Synchronization:** Field switches and camera angle changes complicate the automated detection of team-side transitions.
- **Tracking Resets:** These broadcast-specific artifacts directly motivated the implementation of the *Spatial Jump Threshold* (τ) described in Section 3.5.2, which allows the system to detect these non-physical shifts and reset the tracking state to prevent the accumulation of erroneous trajectories.

Despite these limitations, the dataset serves as a robust benchmark for evaluating detection performance and short-term trajectory reconstruction within individual rallies.

4.2 Model setup

The experimental phase could be divided into two phases: first is the evaluation of pre-trained models and then the evaluation of the fine-tuned version.

Each configuration is evaluated at three different `imgsz` value: 640 pixels (default one), 960 and 1280. It is expected that increasing the image size will lead to improved detection performance.

The performance of all model-image size configurations is evaluated using two metrics:

- **Precision** (P): fraction of predicted positives that are correct
$$P = \frac{TP}{TP + FP}$$
where TP are true positives and FP are false positives.
- **Recall** (R): fraction of ground-truth objects that are successfully detected
$$R = \frac{TP}{TP + FN}$$
where FN are false negatives.

4.3 Model results

We now report the performance of our approach across all configurations. As shown in Table 4, the results for person detection are already quite good (precision and recall > 82%), whereas the ball class shows significantly lower values, especially for recall (ranging from 0% to 4.8%).

After fine-tuning for 20 epochs, as reported in Table 5, the performance improves substantially:

- For the person class, both precision and recall are nearly perfect, with a minimum value of 96.8%.
- For the ball class, results increase considerably, particularly in terms of recall.

Among all configurations, the best performing configuration is YOLO10n with an image size of 1280, as it provides the best trade-off between person and ball detection performance. In particular, the recall for the ball class achives a remarkable improvement, with an increase of 54.7% compared to the pre-trained model (see Table 3).

YOLO10n imgsz=1280			
metric	**class**	Δ	**avg**Δ
precision	*person*	+13.2%	+7.0%
	sport ball	+0.7%	
recall	*person*	+6.6%	+30.8%
	sport ball	+54.7%	

Table 3: Improvement (Δ) of the fine-tuned model compared to the pre-trained one

4.4 Statistics Results

Evaluating the accuracy of derived tactical statistics presents a unique challenge, as there is no existing "ground truth" dataset for the projected 2D coordinates of padel players on a meter-scale court. Instead, the quality of the output is assessed through spatial plausibility and the *Estimation Ratio* (E_r). As illustrated in Figure 3, the high-density areas in the heatmap directly correspond to the players' positions: this alignment confirms that the homography matrix successfully transforms pixel-level movement into a coherent top-down representation.

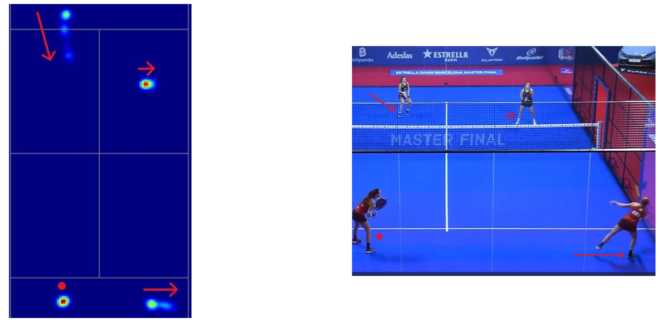

Figure 3: Alignment between source video and generated heatmap showing spatial consistency at the baseline.

Estimation Ratio Analysis: In a test case involving the first 3,145 frames of a female match, the system yielded a remarkably low $E_r \approx 0.058\%$. This indicates that for 99.94% of the duration, player positions were derived directly from real YOLO detections, with "Ghost Points" only being generated for a negligible number of frames.

5 Conclusion

The experimental evaluation shows that the approach is effective for player detection, providing stable performance across different model configurations and input resolutions. The resulting spatial visualizations confirm the potential of detection-based analytics to capture meaningful patterns of player positioning and movement, demonstrating applicability in real-world padel analysis scenarios.

Several challenges were identified during evaluation. Ball detection remains significantly more difficult than player detection due to the small object size, high speed, and frequent occlusions, leading to limited recall. Detection performance was also found to be sensitive to input resolution, highlighting a clear trade-off between accuracy and computational efficiency. Moreover, missing or noisy detections at the frame level may affect the reliability of aggregated statistics, underscoring the importance of robust post-processing and temporal consistency mechanisms.

Future developments will focus on improving ball detection through fine-tuning strategies, data augmentation, and tracking-based refinement. Additional directions include more robust trajectory estimation, shot-level event detection, and extensions toward multi-camera configurations. These improvements would further enhance the reliability and analytical depth of automated padel performance analysis systems.

Overall, the results provide a solid experimental baseline and outline clear directions toward scalable and data-driven padel analytics based on computer vision.

References

[1] Amazon Prime Video. 2025. *Prime Vision: Bringing Advanced Analytics to Live Sports Broadcasts.* https://www.advanced-television.com/2025/09/16/prime-video-bring-prime-vision-to-champions-league/

[2] Roberto Bada-Nerín, Paula Rodríguez, Denis Kreibel, Joao Víctor Teodoro Rodríguez, Oscar D. Pedrayes, Ruben Usamentiaga, Yago Fontenla-Seco, Pablo Ben Lestón, Sebastián Rodríguez Trillo, Nicolás Lozano García, and Franco Mosquera Bonasorte. 2025. *PadelTracker100: A Dataset for Intelligent Player and Ball Tracking in Padel Sports.* doi:10.5281/zenodo.17020011

[3] DataCamp. 2024. *YOLO Object Detection Explained.* https://www.datacamp.com/blog/yolo-object-detection-explained

[4] Martin A Fischler and Robert C Bolles. 1981. Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Commun. ACM* (1981).

[5] Ultralytics. 2024. Train settings. https://docs.ultralytics.com/it/usage/cfg/#train-settings.

[6] Ultralytics. 2024. YOLO11 Models. https://docs.ultralytics.com/it/models/yolo11/.

[7] Ultralytics. 2024. YOLOv10 Models. https://docs.ultralytics.com/it/models/yolov10/.

[8] Ultralytics. 2024. YOLOv9 Models. https://docs.ultralytics.com/it/models/yolov9/.

model	metric	class	imgsz 640		imgsz 960		imgsz 1280	
yolo11n	*precision*	person	*88.4%*	47.8%	87.5%	65.5%	86.5%	73.2%
		sports ball	7.2%		43.6%		59.8%	
	recall	person	85.0%	42.5%	88.7%	44.8%	92.0%	**48.4%**
		sports ball	0.0%		0.9%		*4.8%*	
yolo10n	*precision*	person	86.5%	54.5%	88.3%	74.4%	85.7%	**75.6%**
		sports ball	22.6%		60.5%		*65.6%*	
	recall	person	88.1%	44.1%	91.5%	46.5%	*93.0%*	48.2%
		sports ball	0.2%		1.4%		3.5%	
yolo9t	*precision*	person	85.2%	43.4%	87.8%	56.9%	85.2%	70.7%
		sports ball	1.6%		25.9%		56.1%	
	recall	person	81.9%	41.0%	85.2%	42.8%	87.7%	45.0%
		sports ball	0.0%		0.3%		2.2%	

Table 4: pre-train evaluation

model	metric	class	imgsz 640		imgsz 960		imgsz 1280	
yolo11n	*precision*	person	99.5%	74.5%	*99.8%*	79.6%	99.7%	82.9%
		sports ball	49.6%		59.4%		66.1%	
	recall	person	99.8%	54.5%	*99.9%*	65.9%	99.9%	76.3%
		sports ball	9.1%		32.0%		52.7%	
yolo10n	*precision*	person	97.9%	70.1%	96.8%	80.2%	98.9%	82.6%
		sports ball	42.4%		63.5%		66.3%	
	recall	person	99.8%	55.1%	99.6%	68.6%	99.6%	**79.0%**
		sports ball	10.3%		37.5%		*58.2%*	
yolo9t	*precision*	person	98.2%	58.7%	99.7%	82.6%	99.8%	**84.8%**
		sports ball	19.3%		65.5%		*69.8%*	
	recall	person	99.9%	58.0%	99.9%	66.2%	99.9%	74.3%
		sports ball	16.1%		32.6%		48.8%	

Table 5: fine-tuning evaluation